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# A Novel Approach to Farm Weather Prediction with Hybrid CNN, LSTM, and Attention Mechanisms

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*Abstract*— **This paper presents a comprehensive approach for forecasting weather parameters such as temperature, humidity, and wind speed, using ML and DL techniques. The dataset with weather data covering several years, with features including date, temperature, wind speed, humidity, and precipitation is collected. The data pre-processing involved converting the 'Date' column to a datetime format and extracting key features such as 'Month', 'Season', and 'Day' to capture seasonal and temporal patterns. These features were used for further analysis and modeling. Exploratory data analysis was then performed to visualize seasonal patterns, followed by correlation and trend analyses. For predictive modeling, a range of ML models were applied, including Random Forest Regressor, SVM, XGB, and KNN. These models were evaluated based on several performance metrics. To further enhance the predictive performance, a DL model LSTM applied. Later, novel hybrid models combining CNN with LSTM and CNN+LSTM+Attention architecture demonstrated significant improvements in performance. The hybrid models showed superior generalization capabilities and accuracy in predicting weather parameters compared to traditional models.**

*Keywords*— Weather prediction, Deep Learning, hybrid models, CNN, LSTM, Attention mechanism.

## I. INTRODUCTION

Weather forecasting is an essential aspect of various industries like agriculture, transportation, energy, and disaster management. Correct weather forecast of weather parameters like temperature, humidity, wind speed, and precipitation is crucial because it provides valuable information for effective decision-making, eliminates hazards arising from extreme weather phenomena. For example, in agriculture, these predictions can have a significant impact on crop management practices, irrigation schedules, pest control, and overall yield prediction. Accurate and efficient weather forecasting, therefore provides significant benefits to food stockpilers, resource economists, and policy makers alike.

Traditional weather forecasting methods depend on Numerical Weather Prediction (NWP) models that use physical equations to describe the underlying dynamics of atmospheric processes. Although NWP models have been found to be quite accurate, their computational intensity and the expertise needed to interpret their results are still a liability. Recently, ML and DL methods have been presented as a strong approach for weather forecasting. These data driven methods learn complex relationships from a ton of data, and have the potential to create more accurate and computationally efficient predictions of the weather.

While ML and DL have improved many Weather Prediction problems, finding the best approach depends on the weather system being considered, as well as the future time period in which each model is present. Especially, the accuracy of the predictions can be restrained because of how brilliantly the models predict the generalization in unseen data and their sensitivity against noisy or incomplete input data. New hybrid solutions integrating various ML and DL algorithms are thus proposed to deal with these issues. These hybrid models combine the advantages of various techniques, which allows them to extract features, recognize patterns, and make predictions more accurately.

In this work, a new method proposed where CNN, LSTM networks and Attention mechanism are combined in a hybrid model for weather prediction. The CNN part of the network is used to extract features from each input frame, and based on the learned features, the LSTM network captures long range temporal dependencies in the data. Attention mechanism is also applied to capture the final characteristics and improve prediction accuracy. This hybrid model evaluated on a data set incorporating the weather factors such as temperature, wind speed, humidity, and precipitation over years. The combination of CNN + LSTM + Attention better formed and given more robust results than traditional ML outputs results.

## II. LITERATURE SURVEY

In [1], the authors investigated data-driven approaches for weather and climate forecasting, and recognized their potential for addressing NWP constraints such as high computation cost and uncertainty. The NWP (most variables) was outperformed by data-driven models trained on reanalysis data and forecast times were reduced. While having its advantages, interpretability and uncertainty evaluation are still challenges. Future work may incorporate larger models, physical limitations, and evaluation methods. In [2], the authors used CNNs to refine long-range temperature forecasts by analysing geopotential heights from different tropospheric layers. They evaluated extreme weather events such as the summer heatwave that swept across the United States in 2012 or the cold winter of 2014. In [3], the authors presented ML algorithms, such as decision trees, SVM, neural networks, etc., to enhance short-term weather forecasting. Based on historical data and real-time parameters as inputs, they studied model performance and established the ability of ML to improve forecast accuracy and assist decision-making.

Using historical information and real-time inputs for assessment and improvement of prediction accuracy, the authors of [4] implemented ML algorithms such as decision trees, SVM, and neural networks for short-term weather prediction. The authors of [5] examined weather prediction by ML techniques on a Kaggle dataset based on Seattle. They tried KNN, SVM, gradient boosting, XGBoost, logistic regression, and random forest, with gradient boosting obtaining 80.95% accuracy. They emphasized ML's capability in recognizing non-linear relationships in intricate weather datasets, making it easier and more convenient for meteorologists to enhance prediction accuracy and account for uncertainties in atmospheric dynamics.

In [6], the authors identified some problems with current weather prediction technologies, and in order to overcome these challenges, they employed ML algorithms to eliminate multicollinearity and outliers in historical data records of closed-loop data of weather parameters and factors. Using the optimal ML techniques available, they created a model that would predict future weather based upon the historical data that was available.

Inspired by the DLWP-CS model, the authors in [7] introduced a modified DL technique MDLWP-CS to improve weather prediction with cubed spheres. They also pointed out the challenge of predicting precipitation is a nonlinear domain and turned the architecture into a spatio-temporal mapping space. Bi-LSTM and RNN used by the authors of [8] for weather prediction with historical data points measured in two locations in London. The first model achieved 73% accuracy for the Kew Gardens temperature and, while the second predicted 72-hour temperature and humidity values at Heathrow.

In [9], the authors reviewed 500 peer-reviewed research articles published from 2018 in the field of ML techniques applied in climate and numerical weather forecasting. Key research topics detected were photovoltaic energy, wind energy, atmospheric processes and climate change. Prediction of weather for sparse regions in India where no weather stations are located [10]. They adopted different ML techniques by extracting data such as temperature, humidity, wind speed, and cloud cover.

According to [11], proper weather forecasting is essential in many activities, such as agriculture, and the growing effects of climate change indicate an improvement in weather prediction models. They pointed out that preventing natural disasters is not enough on its own, and something as simple as regular weather reports cannot do it, so advanced information and methods of analyzing statistical data all must be included.

In [12], the authors investigated the possibility of forgoing numerical weather models entirely (at least the traditional formulations) and using DL to forecast the weather instead. To tackle the challenges arising from the non-linear relationship between input weather parameters and output weather condition the authors of [13] proposed a neural network-based model for weather prediction and reported good results.

## III. RESEARCH METHODOLOGY

Figure 1 shows the proposed methodology.

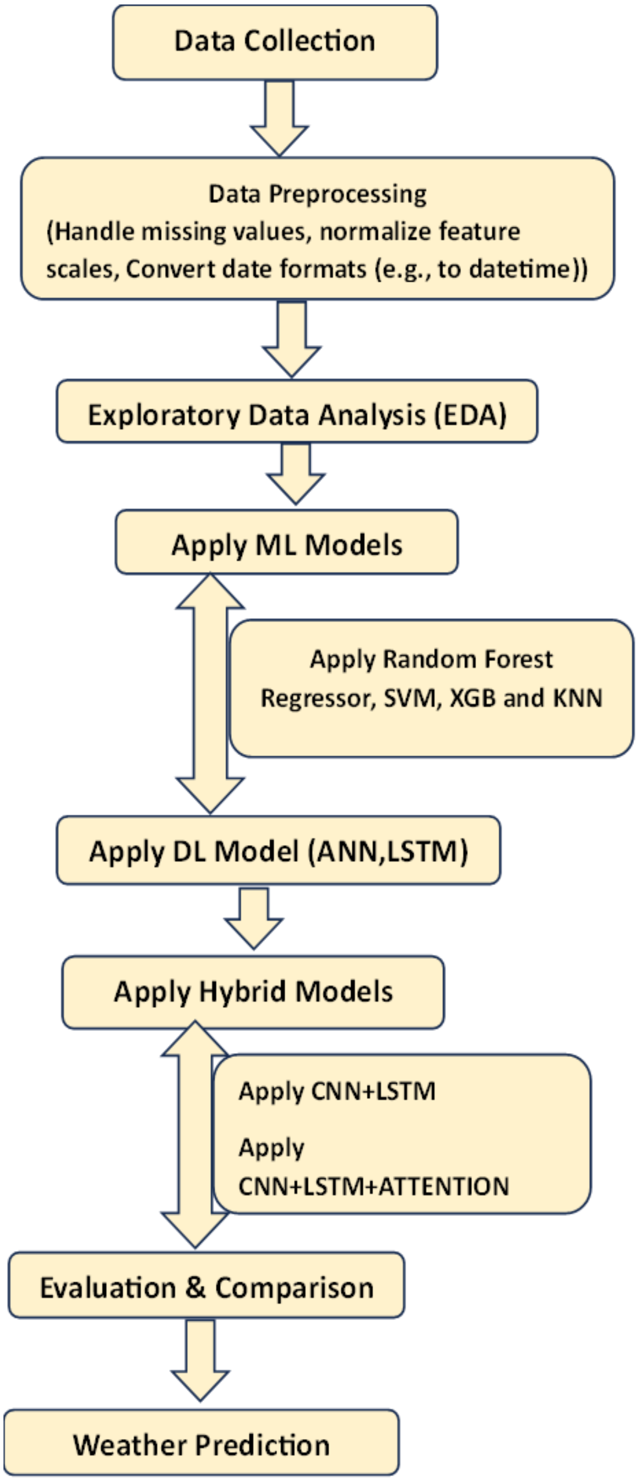


Fig. 1. Proposed Method

The hybrid approach for weather forecasting combines ML and DL models to enhance the prediction of weather statistics like temperature, humidity, wind speed, and precipitation. The methodology involves the following several steps. After loading the raw dataset, we performed data cleaning procedures, missing value treatment, normalized our features scales, and extracted some key temporal features season, month, day of the week to capture any seasonal effects. Then features were extracted for analysis and modeling.

First, several traditional ML models, including random forest regressor, SVM, XGB and KNN applied on the data set as a benchmark for performance. The measures used are R squared, MAE and MSE. These conventional ML models, although might show some predictive power, they can't satisfy accurate forecasts and a deeper level of analysis is necessary.

In the next phase, DL models ANN/LSTM models used for training to model both complex temporal dependencies and long-term trends in the weather data, a long-standing challenge for more traditional ML models. LSTM performed well with good results. Later, a hybrid CNN+LSTM was proposed that combines convolutional layers along with LSTM allowing better extraction of spatial features leading to better prediction accuracy. To further increase performance, an attention mechanism was incorporated into the CNN+LSTM architecture. It allowed the model to focus on the most relevant temporal features, enhancing its capacity to handle long sequences of weather data and improve forecasting accuracy.

## IV. EXPERIMENTS AND RESULTS

### A. Data Collection

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The weather data accumulated for this analysis originated from Kaggle [14] containing daily weather data in Unagatla village situated in A.P, India, spanning the period from 08-01-2006 to 03-02-2023. It holds 6,883 entries. The dataset contains various weather-related parameters such as temperature, humidity, wind speed, and precipitation.

### B. Data Preprocessing

The data preprocessing starts by changing the 'Date' column to a datetime format, enabling the derivation of meaningful time-based features. The 'Date' column is altered to capture specific temporal patterns, which is vital for comprehending the seasonality and trends in weather information. Crucial features including 'Month', 'Season', and 'DayOfYear' are then extracted from the 'Date' column. The 'Month' feature is computed to recognize the specific month of each weather record, which plays a pivotal role in examining seasonal variations. The 'Season' feature is formed by mapping each month to the matching season such as Winter, Spring, Summer or Fall to capture the impacts of diverse seasons on weather tendencies. The 'DayOfYear' feature is additionally calculated to represent the day of the year for every data point, offering insight into the annual weather cycle. Some records have dates that are harder to transform requiring more nuanced processing approaches. The extraction of features uncovers intricate periodic weather designs underpinning the source information.

### C. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to better understand the seasonal patterns and distributions of key weather parameters such as temperature, humidity, wind speed, and precipitation. Visualizations are created to identify trends and anomalies that could help in building predictive models. Boxplots are generated for each weather parameter against the 'Season' feature. This allows for a clear comparison of how each parameter varies across different seasons.

From figure 2, it is observed that, For MaxT, a clear seasonal pattern is observed, with the highest temperatures occurring in summer and the lowest in winter. The spread, or interquartile range (IQR), remains relatively consistent across seasons, indicating similar variability in temperatures within each season.

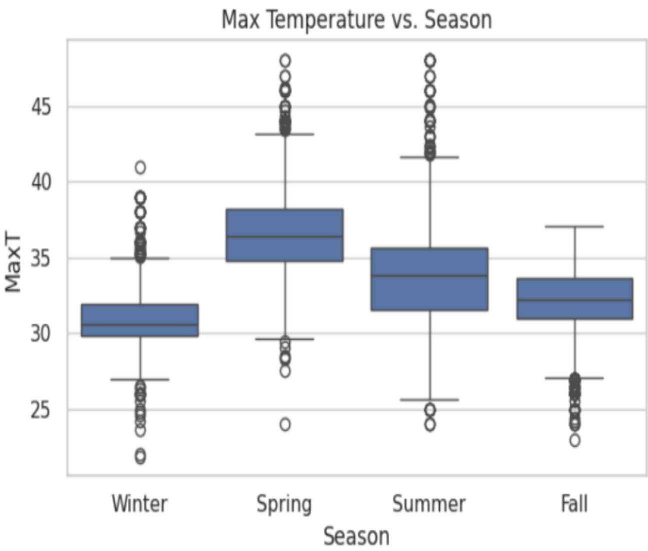


Fig. 2. Boxplot of Maximum Temperature across Seasons

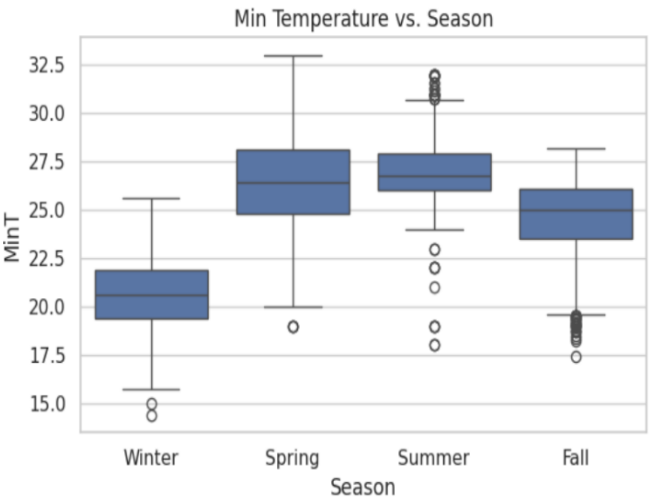


Fig. 3. Boxplot of Minimum Temperature across Seasons

From figure 3, it is observed that MinT also displays a strong seasonal pattern, mirroring MaxT, with the lowest temperatures in winter and the highest in summer. However, the spread of MinT is slightly wider in winter, suggesting greater variability in minimum temperatures during this season. From figure 4, it is observed that the Wind Speed shows a slight tendency for higher values in spring and summer, although the overall seasonal variation is not as pronounced as for temperature. The spread of wind speed is relatively consistent across seasons, indicating similar variability in wind speeds throughout the year.

From figure 5, it is observed that the Humidity demonstrates a clear seasonal pattern, with higher humidity values observed in summer and lower values in winter. The spread of humidity is slightly wider in summer, indicating greater variability during this season.

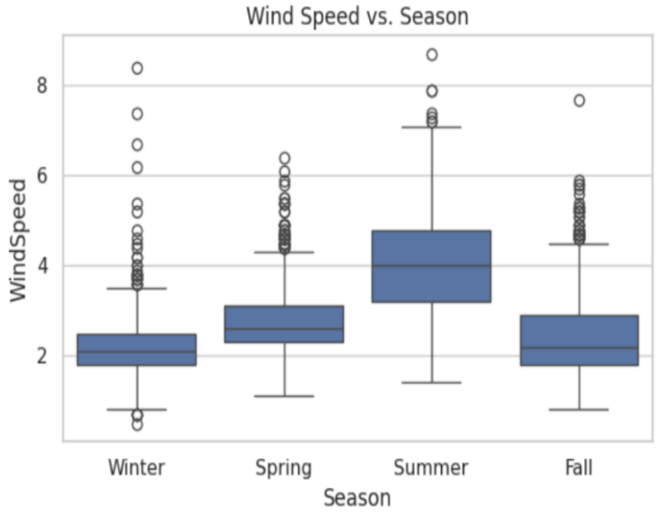


Fig. 4. Boxplot of Wind Speed across Seasons

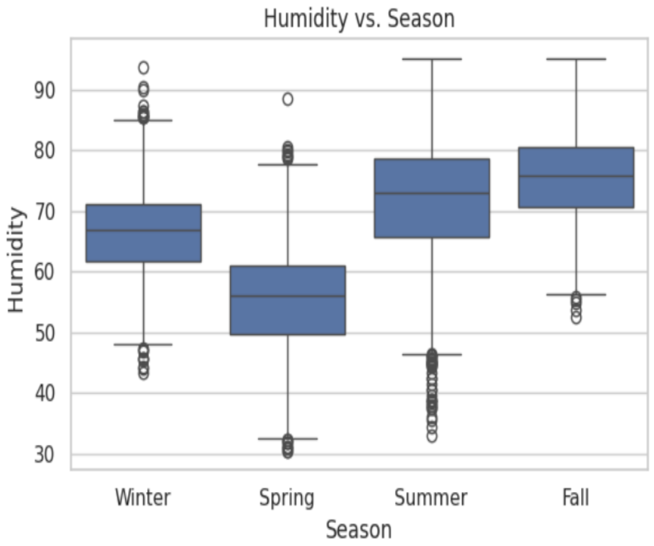


Fig. 5. Boxplot of Humidity across Seasons

From figure 6, it is observed that the Precipitation follows a distinct seasonal pattern, exhibiting higher values in summer and lower values in winter. The spread is notably wider in summer, suggesting greater variability in precipitation during this time.

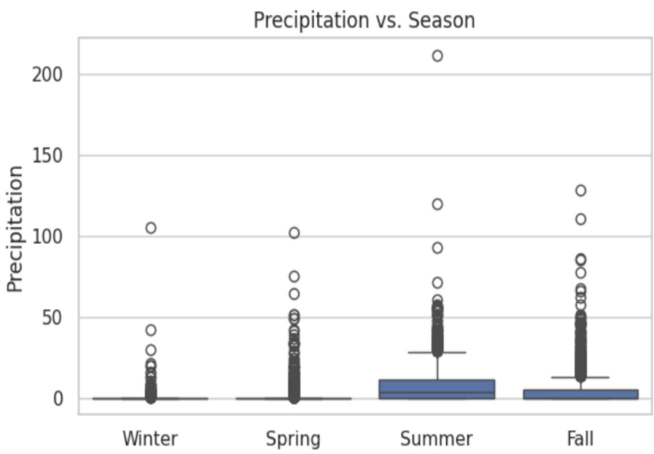


Fig. 6. Boxplot of Precipitation across Seasons

### D. Seasonal Analysis of Weather Parameters

Figure 7 shows the seasonal analysis of key weather parameters, revealing important trends. The highest average maximum temperature (MaxT) is observed in Summer, followed by Fall, Spring, and Winter, aligning with the expected seasonal pattern of warmer temperatures in summer and cooler temperatures in winter. Similarly, the highest average minimum temperature (MinT) occurs in Summer, with the lowest in Winter. Wind speed shows a moderate seasonal variation, with slightly higher averages in Spring and Summer compared to Fall and Winter. Humidity follows a distinct seasonal pattern, with the highest average humidity in Summer and the lowest in Spring, consistent with the general understanding that humidity tends to be higher in warmer and wetter seasons. Precipitation also displays a clear seasonal trend, with the highest average precipitation in Summer and the lowest in Spring, indicating more rainfall during the summer months. These observations provide valuable insights into the seasonal weather variations of the region, which can enhance the understanding and prediction of local weather patterns.

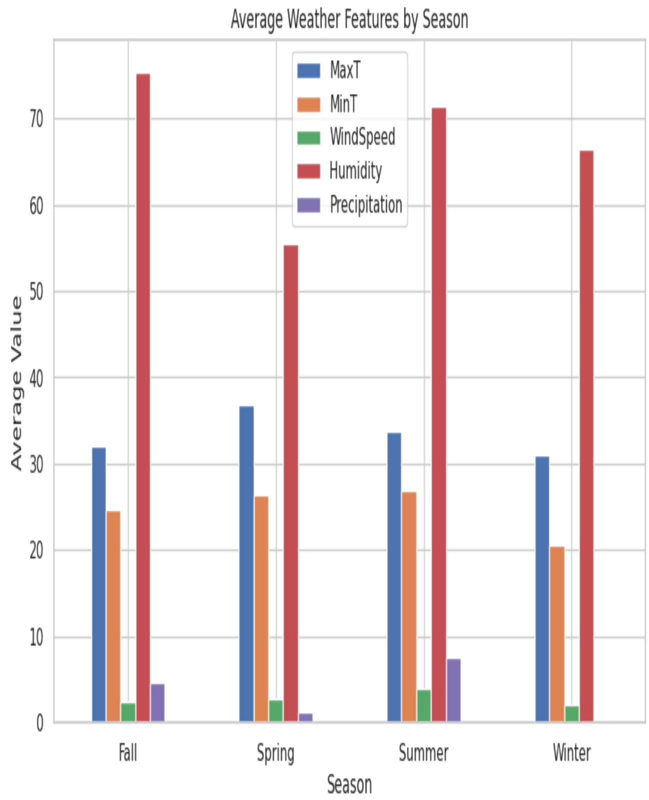


Fig. 7. Boxplot of Precipitation across Seasons

### E. Apply ML models

Table I and figure 8 shows the performance of four ML models. Random Forest achieves the highest $R^2$ of 0.81 for MaxT, with consistent performance, although $R^2$ drops to 0.73 and 0.70 for Humidity and Precipitation. SVR performs reasonably well with an $R^2$ of 0.75 for MaxT but drops to 0.67 and 0.65 for Humidity and Precipitation. XGBoost shows stable performance with $R^2$ values around 0.73. KNN performs the weakest, especially for MinT (0.50) and WindSpeed (0.55).

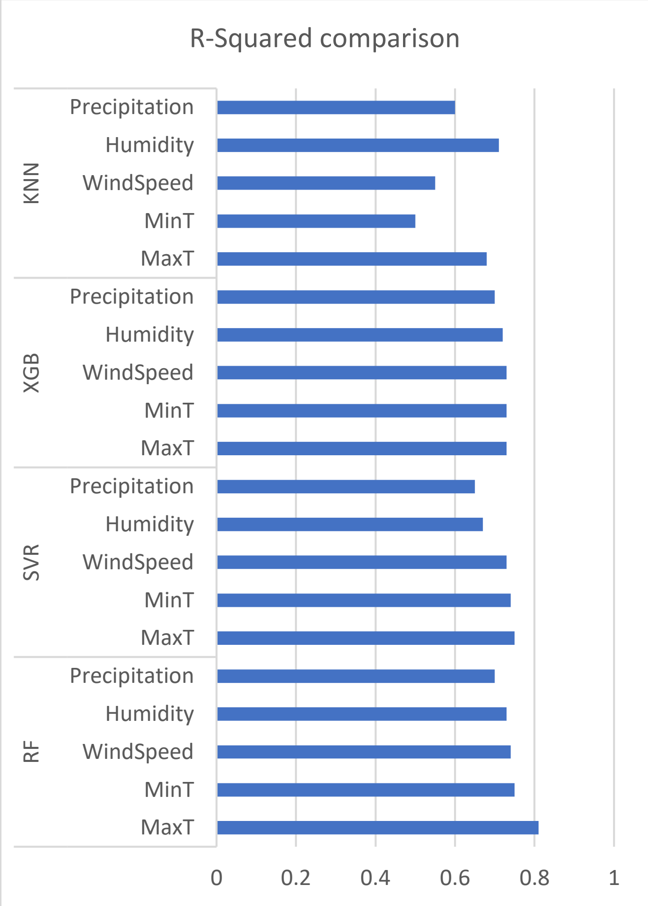


Fig. 8. R-Squared Comparison of ML models

TABLE I. RESULTS WITH ML MODELS

| Model | Feature | MAE | MSE | RMSE | R2 |
|---|---|---|---|---|---|
| Random Forest | MaxT | 1.39 | 3.48 | 1.86 | 0.81 |
| | MinT | 0.52 | 0.50 | 0.71 | 0.75 |
| | WindSpeed | 0.52 | 0.50 | 0.71 | 0.74 |
| | Humidity | 4.64 | 35.64 | 5.97 | 0.73 |
| | Precipitation | 3.30 | 49.52 | 7.04 | 0.70 |
| SVR | MaxT | 2.02 | 6.71 | 2.59 | 0.75 |
| | MinT | 0.64 | 0.77 | 0.88 | 0.74 |
| | WindSpeed | 0.64 | 0.77 | 0.88 | 0.73 |
| | Humidity | 5.56 | 50.94 | 7.14 | 0.67 |
| | Precipitation | 3.15 | 73.57 | 8.58 | 0.65 |
| XGBoost | MaxT | 1.37 | 3.33 | 1.83 | 0.73 |
| | MinT | 0.53 | 0.51 | 0.72 | 0.73 |
| | WindSpeed | 0.53 | 0.51 | 0.72 | 0.73 |
| | Humidity | 4.62 | 35.84 | 5.99 | 0.72 |
| | Precipitation | 3.37 | 51.36 | 7.17 | 0.70 |
| KNN | MaxT | 1.44 | 3.86 | 1.96 | 0.68 |
| | MinT | 0.57 | 0.60 | 0.77 | 0.50 |
| | WindSpeed | 0.57 | 0.60 | 0.77 | 0.55 |
| | Humidity | 4.80 | 37.55 | 6.13 | 0.71 |
| | Precipitation | 3.24 | 54.13 | 7.36 | 0.60 |

### F. *Applying DL Models*

In the next step, ANN and LSTM models were applied to analyze the data. The ANN model, with a three-layer architecture (64, 32, and 16 neurons), demonstrated good performance, achieving an $R^2$ value of 0.85 for MaxT, and 0.81, 0.82, 0.80, and 0.80 for MinT, WindSpeed, Humidity, and Precipitation, respectively. In comparison, the LSTM model, designed with two LSTM layers followed by dense layers, outperformed the ANN model. It achieved an $R^2$ value of 0.87 for MaxT, with corresponding values of 0.86, 0.83, 0.82, and 0.84 for MinT, WindSpeed, Humidity, and Precipitation, respectively. The results are depicted in figure 9 and Table II.

TABLE II. RESULTS WITH DL MODELS

| Model | Feature | R-Squared |
|---|---|---|
| ANN | MaxT | 0.85 |
| | MinT | 0.81 |
| | WindSpeed | 0.82 |
| | Humidity | 0.80 |
| | Precipitation | 0.80 |
| LSTM | MaxT | 0.87 |
| | MinT | 0.86 |
| | WindSpeed | 0.83 |
| | Humidity | 0.82 |
| | Precipitation | 0.84 |

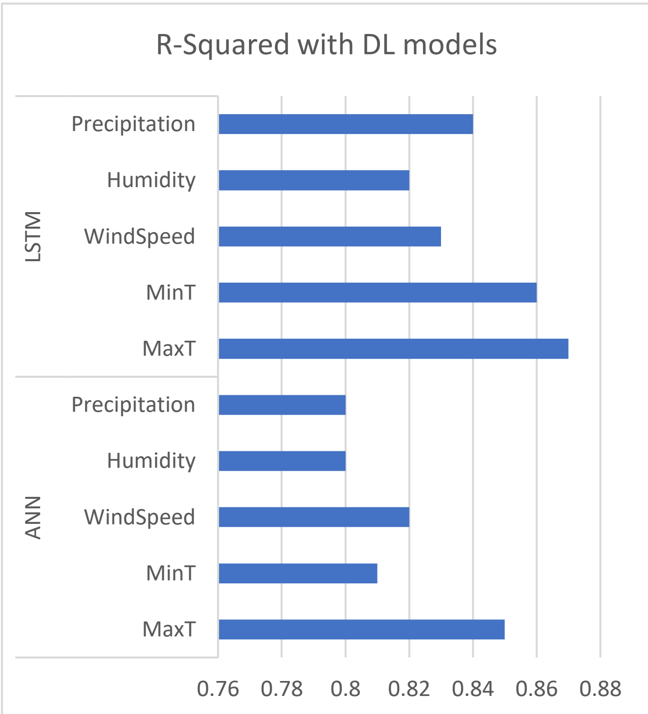


Fig. 9. R-squared values with DL models

### G. *Applying CNN+LSTM Hybrid Model*

To further enhance the predictive accuracy of weather parameter forecasting, hybrid models combining the strengths of CNNs, LSTMs and attention mechanisms are applied. The hybrid CNN-LSTM model uses the feature extraction capability of CNN and the sequential modeling power of LSTM networks. Initially, a 1D convolutional layer is used with 64 filters and a kernel size of 3 to capture local patterns in time-series data, followed by a max-pooling layer to reduce dimensionality. The extracted features are then flattened and passed to an LSTM layer, which processes sequential dependencies in the data. It uses MSE as the loss function, and MAE as the performance metric. The R squared value achieved with this model is 0.91,0.88,0.87,.86,0.83 for MaxT, MinT, windspeed, humidity and Precipitation respectively.

### *H. Applying CNN+LSTM +Attention Hybrid Model*

The CNN+LSTM+Attention hybrid model combines convolutional layers for spatial feature extraction, LSTM for temporal dependencies, and an attention mechanism to enhance feature importance. The model processes scaled and reshaped input data, with the attention layer refining the LSTM output for better predictions.

Table III and Figure 10 compare the R-squared performance of CNN+LSTM and CNN+LSTM+Attention models across five features: MaxT, MinT, WindSpeed, Humidity, and Precipitation. Both models perform similarly, with CNN+LSTM achieving slightly higher R-squared for MaxT (0.91 vs. 0.93). However, CNN+LSTM+Attention shows improvement in MinT (0.89 vs. 0.88) and maintains comparable performance for other features. This highlights the attention mechanism's role in enhancing predictions for specific features. Figure 11 shows epoch-wise MSE values with the CNN+LSTM+Attention model.

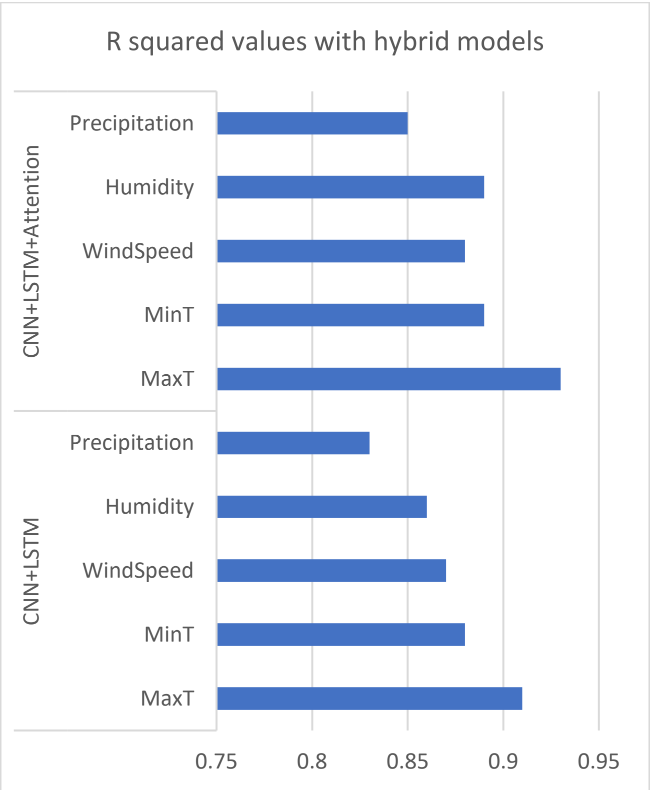


Fig. 10. R-squared values with hybrid models

TABLE III. RESULTS WITH HYBRID MODELS

| Model | Feature | R-Squared |
|---|---|---|
| CNN+LSTM | MaxT | 0.91 |
| | MinT | 0.88 |
| | WindSpee | 0.87 |
| | Humidity | 0.86 |
| | Precipitati | 0.83 |
| CNN+LSTM+Attention | MaxT | 0.93 |
| | MinT | 0.89 |
| | WindSpee | 0.88 |
| | Humidity | 0.89 |
| | Precipitati | 0.85 |

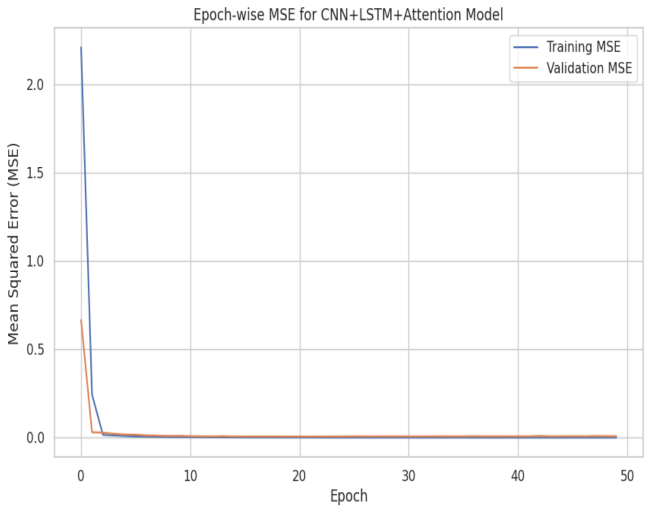


Fig. 11. Epoch wise MSE for CNN+LSTM+Attention with MaxT target

## V. CONCLUSION

This paper explored various ML and DL models for forecasting weather parameters like temperature, humidity, and wind speed. Traditional ML models applied and Random Forest, showed good performance, particularly for MaxT, but struggled with humidity and precipitation. Deep learning models like ANN and LSTM performed reasonably well but were outperformed by hybrid models. The CNN+LSTM and CNN+LSTM+Attention models delivered superior perofrmance and generalization, especially for MaxT, MinT, and WindSpeed. Overall, hybrid DL models proved to be more effective than traditional ML models for weather forecasting, offering enhanced predictive accuracy and robustness.

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